

A cut-cell Cartesian method for linear and causal-viscous oscillations of relativistic stars in the Cowling approximation

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We present a 3+1D time-domain method for the linear oscillations of a relativistic star on a frozen Tolman–Oppenheimer–Volkoff background (the Cowling approximation), on a Cartesian grid that does not conform to the stellar surface. The linearized fluid equations are evolved in a covariant-momentum form in which the factor $(\varepsilon_0 + p_0)$ cancels from every right-hand side, removing the surface singularity of the velocity formulation; the lapse multiplying the Euler equation is derived from $\nabla_\mu \delta T^\mu{}_\nu = 0$ and shown to be essential, its omission producing a resolution-independent +5% frequency error. The stellar surface is treated by cut cells: the volume fraction and the six face apertures of every surface cell reduce to one-dimensional Gauss quadratures of a single disc-rectangle area, and the free-surface condition ($\Delta p = 0$, no mass or viscous flux through the moving boundary) is imposed at the true sphere $r = R$ by simply omitting the boundary face from the aperture-weighted finite-volume divergence. Replacing the conventional stair-stepped mask, which imposes a rigid wall, by this treatment moves the $\ell = 2$ f -mode of the benchmark $\Gamma = 2$ polytrope from -6.95% to -0.08% of the frequency-domain shooting value at $N = 48$ cells across the box, with p_1 to -1% ; three smooth barotropes agree to $\leq 0.08\%$. With the cut-cell geometry made exact (every quadrature split at its kinks and symmetrized over the axes), the engine seeds any real $Y_{\ell m}$ on the full cube: the E_g partners Y_{20} and $\text{Re } Y_{22}$ give frequencies identical to fourteen significant figures, as the cubic group requires, while the T_{2g} member $\text{Re } Y_{21}$ sits 1.13% lower at $N = 24$ —a direct measurement of the grid’s cubic anisotropy. We extend the method to Bemfica–Disconzi–Noronha–Kovtun (BDNK) first-order causal viscous hydrodynamics, keeping the frame relaxation times so that the linearized system remains hyperbolic, with the covariant expansion and the lapse in both recovery equations. In the three hydrodynamic frames of Clarisse *et al.* the viscous damping rate of the f -mode is frame-invariant to 0.045% and converges to 99% of the Navier–Stokes dissipation integral; the viscous frequency shift, once the Kreiss–Oliger artifact that dominated it is removed, converges to the parameter-free damped-oscillator value $-(\gamma^2 - \gamma_0^2)/2\omega^2$ (-0.125% measured against -0.133% predicted at $N = 48$); while the linear stability bound $\tau_Q w_0 c_s^2 > \frac{4}{3}\eta$ is shown to be sufficient but not necessary: the marginally causal frame violates it in every cell yet evolves stably for 46 periods, because the unstable band sits above a critical wavenumber the stencil cannot resolve, and a frame violating it in half the star destabilizes first. We report the method’s limits with equal weight: realistic crusted equations of state excite a growing surface mode for $N \geq 40$ whose discriminant is the surface index n in $\varepsilon \propto (R - r)^n$; the Kreiss–Oliger coefficient is a $\pm 0.25\%$ systematic on absolute frequencies and the dominant one on viscous frequency shifts; and time-domain frequency resolution at affordable cost is 4% of f , so every sub-percent figure is a parametric estimate.

I. INTRODUCTION

Frequency-domain eigensolvers on spherically symmetric backgrounds compute neutron-star oscillation modes to five or six figures and remain the standard of accuracy. A time-domain 3+1D evolution is worse at that task by construction: its spectral resolution is $1/T$, its surface is not aligned with the grid, and its dissipation is not zero. The case for building one anyway is what it can do that the eigensolvers cannot—carry a non-axisymmetric perturbation, couple modes nonlinearly, and eventually sit inside a dynamical-spacetime merger code—and the precondition for any of that is that the linear, axisymmetric, Cowling problem be reproduced first. This paper is about that precondition.

The difficulty is the surface. A relativistic star’s density and pressure go to zero at $r = R$, so a Cartesian grid intersects a sphere of vanishing enthalpy along a stair-stepped set of cell faces. The common treatments are

an atmosphere (which introduces a spurious low-density fluid), an excised shell (which imposes a wall inside the star), or a binary interior mask (which imposes a wall on the staircase). We show below that the last of these—the most natural choice for a linearized code, since exterior cells simply stay at zero—produces a -7% f -mode frequency error that does *not* converge away between $N = 32$ and $N = 40$, from a run that looks perfectly healthy. The error is not the geometric volume error of the staircase, which is at the 10^{-4} level by $N = 72$; it is the boundary condition. A stationary exterior imposes $\delta S_i = 0$ on the staircase, i.e. a rigid wall, where the physics of the f -mode demands a free surface that moves with the fluid. Cut cells fix the condition, not the geometry.

The second subject is viscosity. The BDNK formulation [1, 2] is the first first-order viscous theory that is causal, linearly stable and strongly hyperbolic, at the price of a set of frame relaxation times $(\tau_\varepsilon, \tau_P, \tau_Q)$ on

which no physical prediction may depend. Clarisse *et al.* [3] have recently tested that frame independence in a flux-conservative 1+1D code by running three distinct frames. We do the same test on a star in 3+1D, and use the stellar setting to probe the linear stability bound that the theory's own analysis provides.

Section II states the equations; Sec. III the discretization, including the cut-cell geometry and the frequency-extraction protocol; Sec. IV validates the ideal engine against a shooting solver and the literature; Sec. V reports the viscous results; Sec. VI compares with a non-linear discontinuous-Galerkin engine on the same problem; Sec. VII records the method's limits and the defects found. Throughout, $G = c = 1$; the benchmark star is the $n = 1$, $K = 100$ polytrope with $\rho_c = 1.28 \times 10^{-3}$, $M = 1.4002 M_\odot$, $R = 14.154$ km, whose relativistic-Cowling $\ell = 2$ spectrum is tabulated in Ref. [4] (Table 3) and reproduced in Ref. [5] (Table VI) as $f, p_1, \dots, p_4 = 1.8825, 4.1060, 6.0298, 7.8670, 9.6663$ kHz.

II. PHYSICAL MODEL

A. Background in areal-Cartesian coordinates

The TOV star $m' = 4\pi r^2 \varepsilon$, $\nu' = 2(m + 4\pi r^3 p)/[r(r - 2m)]$, $p' = -(\varepsilon + p)(m + 4\pi r^3 p)/[r(r - 2m)]$ is integrated by RK4 and embedded in a cube $[-L, L]^3$, $L = 1.2R$, with N cells per dimension. At each cell centre the 1D profile is interpolated in $r = \sqrt{x^2 + y^2 + z^2}$. The areal metric $ds^2 = -e^\nu dt^2 + e^{2\lambda} dr^2 + r^2 d\Omega^2$, $e^{2\lambda} = (1 - 2m/r)^{-1}$, becomes in Cartesian components

$$\gamma_{ij} = \delta_{ij} + (e^{2\lambda} - 1) n_i n_j, \quad \gamma^{ij} = \delta^{ij} + (e^{-2\lambda} - 1) n_i n_j, \quad (1)$$

with $n_i = x_i/r$, $\sqrt{\gamma} = e^\lambda$ and lapse $\alpha = e^{\nu/2}$; the spatial metric has eigenvalues $(e^{2\lambda}, 1, 1)$, a purely radial stretch. Outside $r = R$ the Schwarzschild exterior is used. The local gravity is $\Phi' = \nu'/2 = (m + 4\pi r^3 p)/[r(r - 2m)]$ and $p'_0 = -w_0 \Phi'$, $w_0 \equiv \varepsilon_0 + p_0$.

B. Linearized Cowling equations in momentum form

Evolving the velocity perturbation δv^i fails at the surface: every term carries $1/(\varepsilon_0 + p_0)$. We evolve instead the covariant momentum density $\delta S_i = (\varepsilon_0 + p_0) \delta v_i$, in which that factor cancels identically from every right-hand side (the Valencia momentum form applied to the linear problem). With the barotropic closure $\delta p = c_s^2 \delta \varepsilon$ and $\delta S^i = \gamma^{ij} \delta S_j$,

$$\partial_t \delta \varepsilon = -\frac{1}{\sqrt{\gamma}} \partial_i [\alpha \sqrt{\gamma} \delta S^i] - \alpha \Phi' (n_i \delta S^i), \quad (2)$$

$$\partial_t \delta S_i = -\alpha [\partial_i \delta p + \Phi' n_i (\delta p + \delta \varepsilon)]. \quad (3)$$

The lapse multiplying the whole of Eq. (3) is the load-bearing detail. It follows from $\nabla_\mu \delta T^\mu_\nu = 0$ written as

$\nabla_\mu T^\mu_\nu = (1/\sqrt{-g}) \partial_\mu (\sqrt{-g} T^\mu_\nu) - \frac{1}{2} T^{ab} \partial_\nu g_{ab}$: the flux is $-(1/\sqrt{\gamma}) \partial_i (\alpha \sqrt{\gamma} \delta p)$, the source is $\alpha [-\delta \varepsilon \Phi' + \delta p \partial_i \ln \sqrt{\gamma}]$, and the $\partial_i \ln \sqrt{\gamma}$ pieces cancel between them, leaving a *bare* α on the flat-form right-hand side—not the $\alpha \sqrt{\gamma}$ -inside-the-divergence form, which leaves an uncanceled $\delta p \partial \ln \sqrt{\gamma}$ and overshoots. The $\nu = t$ component of the same identity reproduces Eq. (2) with zero residual, which validates the framework. Without the lapse the continuity and Euler equations carry different time normalizations and every mode frequency acquires a resolution-independent error of $\approx +5\%$. As an independent check, reducing Eqs. (2)–(3) to a single second-order radial ODE and changing dependent variable, $E = \phi(r)V$ with $\phi'/\phi = -[(1 + 3c_s^2)\alpha' + \alpha c_s^2]/(\alpha c_s^2)$, carries them *exactly* onto the McDermott–Van Horn–Scholl (W, V) system of the shooting solver [7]; without the lapse the map does not close. The two codes therefore solve the same continuum problem and any residual is numerical.

The Cowling f -mode is undamped. The physical surface condition is the vanishing of the Lagrangian pressure perturbation, $\Delta p = \delta p + \xi^r p'_0 = 0$; for a polytrope $p'_0(R) = 0$ so $\delta p(R) = 0$, and there is no mass flux and no viscous traction through the moving surface.

C. BDNK causal viscous extension

The general-frame stress tensor [1] is

$$\begin{aligned} T^{\mu\nu} &= (\varepsilon + \mathcal{A}) u^\mu u^\nu + (P + \Pi) \Delta^{\mu\nu} \\ &\quad - 2\eta \sigma^{\mu\nu} + u^\mu \mathcal{Q}^\nu + u^\nu \mathcal{Q}^\mu, \\ \mathcal{A} &= \tau_\varepsilon [D\varepsilon + w\theta], \quad \Pi = -\zeta\theta + \tau_P [D\varepsilon + w\theta], \\ \mathcal{Q}^\nu &= \tau_Q [w D u^\nu + c_s^2 \Delta^{\nu\lambda} \partial_\lambda \varepsilon], \end{aligned} \quad (4)$$

$D = u \cdot \nabla$, $\theta = \nabla \cdot u$, $w = \varepsilon + P$. Linearized about the static star with $u_0^\mu = (1/\alpha, 0, 0, 0)$, the conserved densities carry *proper*-time derivatives of the primitives, $D\delta\varepsilon = \alpha^{-1} \partial_t \delta\varepsilon$, $(Du^i)_{\text{lin}} = \alpha^{-1} \partial_t \delta v^i$, so the recovery is diagonal and each recovered derivative is converted to coordinate time by a factor α :

$$\delta E = \delta\varepsilon + \tau_\varepsilon (\alpha^{-1} \partial_t \delta\varepsilon + w_0 \theta), \quad (5)$$

$$\partial_t \delta\varepsilon = \alpha [(\delta E - \delta\varepsilon)/\tau_\varepsilon - w_0 \theta],$$

$$\delta S^i = w_0 \delta v^i + (\tau_Q w_0 - \eta) \alpha^{-1} \partial_t \delta v^i + \tau_Q c_s^2 \gamma^{il} \partial_l \delta\varepsilon,$$

$$\partial_t \delta v^i = \frac{\alpha (\delta S^i - w_0 \delta v^i - \tau_Q c_s^2 \gamma^{il} \partial_l \delta\varepsilon)}{\tau_Q w_0 - \eta}. \quad (6)$$

Three details are exact on the curved background and each was found by failing without it. The expansion is the four-dimensional one and carries both the lapse and $\sqrt{\gamma}$, $\theta = (\alpha \sqrt{\gamma})^{-1} \partial_i (\alpha \sqrt{\gamma} \delta v^i) = \partial_i \delta v^i + (\Phi' + A'/A)(n \cdot \delta v)$, not the flat divergence. The shear σ^{ij} uses the closed-form Christoffels of $\gamma_{ij} = \delta_{ij} + (A^2 - 1) n_i n_j$, $\Gamma_{bc}^a = n^a [g_1 n_b n_c + g_2 (\delta_{bc} - n_b n_c)]$, verified in all 27 components. And the bulk closure is pinned: $\Pi = c_s^2 \mathcal{A}$ forces $\tau_P = c_s^2 \tau_\varepsilon$, and using $\tau_P = \tau_\varepsilon$ instead reopens a

primitive-feedback instability. The recovery denominator $\tau_Q w_0 - \eta > 0$ is the causality margin (BDN condition (a), $\rho\tau_Q > \eta$). Because the frame times are kept, the linearized system is hyperbolic and the diffusive CFL restriction of the Navier–Stokes limit—unstable for $\hat{\eta} \gtrsim 1$ —does not arise.

A local analysis of the 4×4 longitudinal sound sector ($\delta E, \delta S, \delta \varepsilon, \delta v$) that this system integrates, in flat space with frozen coefficients, gives the linear stability condition

$$\tau_Q w_0 c_s^2 > \frac{4}{3} \eta, \quad (7)$$

not $\tau_Q w_0 > \eta$; it was reproduced to ratio 1.0000 in six $(c_s^2, \hat{\eta})$ combinations and is used in Sec. V.

D. Hydrodynamic frames

Clarisse *et al.* [3] parametrize $\tau_\varepsilon = a_1(1/T)(\eta/s)$, $\tau_Q = a_2(1/T)(\eta/s)$ under $a_1 \geq 4$, $a_2 \geq 3a_1/(a_1 - 1)$, with $F_1 = (25/4, 25/7)$, $F_2 = (25/2, 25/3)$, $F_3 = (25, 25)$ and characteristic speeds $c_+ = 1.00, 0.85, 0.74$. F_1 saturates the second bound exactly, $3(25/4)/(21/4) = 25/7$, which is why $c_+ = 1$: it is the marginally causal frame. At zero chemical potential $w = sT$, so $(1/T)(\eta/s) = \eta/w$ and the stellar map is

$$\tau_\varepsilon(\mathbf{x}) = a_1 \eta(\mathbf{x})/w_0(\mathbf{x}), \quad \tau_Q(\mathbf{x}) = a_2 \eta(\mathbf{x})/w_0(\mathbf{x}), \quad (8)$$

installed cell by cell, with $\eta = \hat{\eta} \varepsilon_0$. Under Eq. (8) the bound (7) collapses to the η -independent $a_2 c_s^2 > 4/3$. The repository reproduces the published c_+ from (a_1, a_2) alone via the Kovtun large- k biquadratic [2]: 1.000000/0.849719/0.741915, differing from the table by two-decimal rounding.

III. NUMERICAL METHOD

A. Discretization

Second-order central differences on the N^3 grid, classical RK4 with $\Delta t = 0.30 \Delta x$ (ideal) or $0.20 \Delta x$ (BDNK), and fourth-derivative Kreiss–Oliger dissipation $-\sigma_{\text{KO}} \sum_d (\dots)$ applied to every evolved field. Exterior cells are never written; interior cells are those with positive volume fraction (cut) or centre inside R (masked). The engine is single-threaded; cost scales as N^4 with a fitted 6.9×10^{-8} s per $N^4 M_\odot$ of evolution, i.e. ~ 900 s for the production unit $N = 48$, $T = 2500 M_\odot$ (23 f -periods).

B. Cut-cell geometry

For each cell whose centre lies within $2\Delta x$ of $r = R$ we need the volume fraction κ inside the sphere and the

six face apertures $a_{x\pm}, a_{y\pm}, a_{z\pm}$ (the fraction of each face inside the sphere). Both reduce to one function: the area of the intersection of a disc of radius a with an axis-aligned rectangle $[y_0, y_1] \times [z_0, z_1]$,

$$\mathcal{D} = \int_{y_-}^{y_+} \ell(y) dy, \quad y_\mp = \max(y_0, -a), \min(y_1, a),$$

$$\ell(y) = [\min(z_1, \zeta) - \max(z_0, -\zeta)]_+, \quad \zeta = \sqrt{a^2 - y^2}, \quad (9)$$

A face at $x = x_f$ meets the sphere in a disc of radius $\sqrt{R^2 - x_f^2}$, so $a_{x\pm} = \mathcal{D}(\sqrt{R^2 - x_f^2}; \dots)/\Delta x^2$ directly; the volume fraction is a second quadrature of $\mathcal{D}$ over x across the cell, $\kappa = \Delta x^{-3} \int \mathcal{D} dx$. No interface-plane reconstruction is needed.

The integrands are only piecewise smooth: $\ell(y)$ has kinks where the disc rim meets a rectangle edge, $\mathcal{D}(a(x))$ has kinks where the rim passes an edge or a corner, and both have a square-root singularity at the rim. A single Gauss rule therefore converges algebraically—24 points left $\sim 10^{-5}$ errors—and because κ integrates over x while the faces integrate over y and z , that error was not symmetric under $x \leftrightarrow y$: the geometry broke the sphere’s own octahedral symmetry at 10^{-5} , which leaked into the $Y_{\ell m}$ moments (Sec. IV B). The implementation is therefore exact: the substitutions $y = a \sin \theta$ and $x = R \sin \varphi$ remove the rim singularities, every integral is split at its kink points ($\cos \theta = \{z_1, -z_0, z_0, -z_1\}/a$; $\cos \varphi = \{|y_{0,1}|, |z_{0,1}|, \text{corner distances}\}/R$), and 16-point Gauss on each smooth piece is exact to round-off; κ is finally averaged over the six axis permutations, under which the true κ is invariant. Verified against the analytic sphere: $\sum \kappa \Delta x^3 = \frac{4}{3} \pi R^3$ to 2×10^{-16} , the x -face apertures equal the transposed y -face apertures to 0.0, and the closure condition—the face-aperture area vectors balancing the boundary face—reproduces $4\pi R^2$ to 8×10^{-4} at $N = 24$ and 4.5×10^{-4} at $N = 32$ (second order). Cells with κ below a floor $\kappa_{\min} = 0.5$ use $\kappa_{\min}$ in the denominator (a small-cell treatment).

C. Aperture-weighted divergences and the free surface

Every spatial derivative in Eqs. (2)–(3) is a divergence or a gradient, and both are evaluated in finite-volume form. For the continuity flux $F^i = \alpha \sqrt{\gamma} \delta S^i$,

$$(\nabla \cdot F)_{ijk} = \frac{1}{\kappa_c \Delta x} \left[a_{x+\frac{1}{2}} (F_i^x + F_{i+1}^x) - a_{x-\frac{1}{2}} (F_{i-1}^x + F_i^x) + (y) + (z) \right], \quad (10)$$

with face values the second-order average of the adjacent cell-centred fluxes. The gradient of δp uses the divergence theorem in the same form, $\int \partial_j \delta p dV = \oint \delta p n_j dA$, so co-ordinate faces contribute $a \delta p$; and the viscous force $\partial_j \pi^{ij}$

likewise. *The boundary face does not appear in any of these sums.* That omission is the free-surface boundary condition: the boundary face would contribute $\delta S^i n_i A_b$ (mass flux), $\delta p_b n_j A_b$ (pressure) and $\pi^{ij} n_j A_b$ (traction), all of which vanish for a free surface that moves with the fluid and carries $\Delta p = 0$. The condition is thereby imposed at the true sphere, and the exterior is never referenced. In the masked engine, by contrast, the central stencil just inside the staircase reads the frozen exterior zeros, imposing $\delta \varepsilon = 0$ (correct, since $\delta p(R) = 0$) and $\delta S_i = 0$ —a rigid wall. This is the whole difference, and Sec. IV quantifies it.

Two remedies that do *not* work are recorded so they are not retried. Zero-gradient extrapolation of δS into an exterior collar with $\delta \varepsilon = 0$ makes the mode collapse (-61% at $N = 32$, -83% at $N = 44$): momentum in a region with no pressure restoring force plus a spurious mass flux in the continuity stencil. And an excised shell (evolving only $w_0 \geq f_{\text{exc}} \max w_0$) merely moves the wall inward, giving $+2.6\%$ at $f_{\text{exc}} = 0.02$ and $+8.2\%$ at 0.08 .

D. Seeding and diagnostics

The seed is a density perturbation $\delta \varepsilon = A \varepsilon_0 (r/R)^\ell Y_{\ell m}(\hat{n})$ with $Y_{\ell m}$ the real solid-harmonic angular factor ($\ell = 2-4$, all m ; cosine type for $m \geq 0$, sine type for $m < 0$) and $A = 10^{-3}$; the radial factor makes $r^\ell Y_{\ell m}$ a polynomial, smooth at the centre. The legacy $\ell = 2$, $m = 0$ seed used $(r/R)^1$ and is retained. The diagnostics are the $\sqrt{\gamma}$ -weighted moments $q_{\ell m}(t) = \sum \delta \varepsilon Y_{\ell m}(\hat{n}) \sqrt{\gamma} \Delta x^3$ over interior cells, any set of which can be recorded in one run. Because the code is linear, amplitude proportionality is exact to 10^{-14} (verified over $\times 100$ and $\times 3000$), which makes the linear engine the control for any nonlinear-engine claim (Sec. VI). The grid is the full cube, not an octant, so every m is admissible.

E. Frequency extraction and its resolution

Two independent estimators are applied to every record and must agree: a direct periodogram $|\sum_n q_n e^{-i\omega t_n}|^2$ with parabolic refinement of the peak, and a matrix pencil (Hankel SVD, 14–16 poles) on the record decimated to ~ 600 samples. Every record is gated before either is applied: finiteness, and the ratio of the envelope at the end to that at the start; a growing record is re-analysed on its largest clean window, and if that window is shorter than ten periods the row is voided rather than reported. A record of length T has spectral resolution $\Delta f = 1/T$, which at the production $T = 2500 M_\odot$ is $0.083 \text{ kHz} = 4.4\%$ of f . *Every sub-percent figure below is therefore a parametric peak estimate, quoted only where the two estimators agree, and no third decimal is meaningful.* Buying record length does not help: at $\sigma_{\text{KO}} = 0.004$, $T = 12000$ actively de-

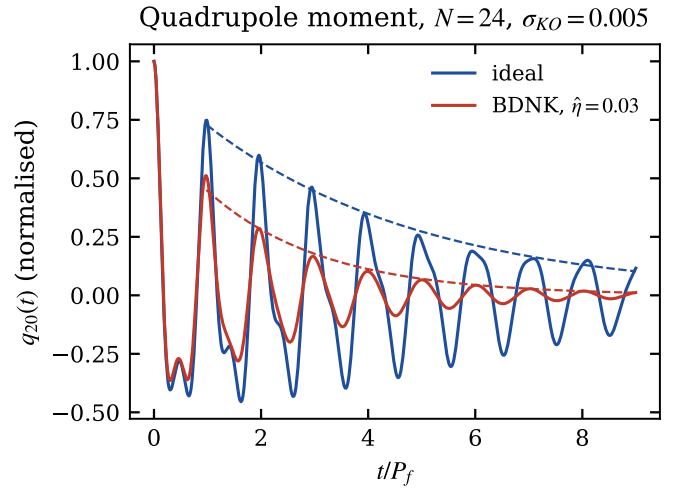


FIG. 1. The $\ell = 2$ quadrupole moment $q_{20}(t)$ of the benchmark star at $N = 24$, $\sigma_{\text{KO}} = 0.005$, for the ideal engine and for BDNK in frame F_1 at $\hat{\eta} = 0.03$, each normalised to its seed. Dashed: exponential fits to the positive peaks. The ideal run decays at the numerical floor γ_0 (Kreiss–Oliger dissipation and the surface); the viscous damping is the difference. The first half-period is the seed relaxing onto the mode.

grades the answer as a late-time surface mode takes over ($-0.30\% \rightarrow -1.67\%$). The damping rate is the slope of a log-linear fit to the local-maximum envelope, returning NaN (not zero) when fewer than three peaks exist.

The viscous damping is measured by a *difference protocol*: every frame is run twice at identical grid and frame times, once at $\hat{\eta}$ and once at $\hat{\eta} = 0$, and $\gamma_{\text{visc}} = \gamma(\hat{\eta}) - \gamma(0)$. The numerical floor $\gamma(0)$ from Kreiss–Oliger and the surface is $1.30 \times 10^{-3} M_\odot^{-1}$ at $N = 40$ – 52% of the viscous rate—so quoting $\gamma(\hat{\eta})$ raw would overstate the damping by $1.5\times$. Figure 1 shows what the engine produces: the ideal run itself decays, at the numerical floor, and the viscous run decays faster; the difference is the physics.

IV. VALIDATION OF THE IDEAL ENGINE

A. The benchmark star

Table I and Fig. 2 are the central validation. With cut cells at $N = 48$ the f -mode is -0.083% from the shooting value and p_1 is -0.98% ; the masked engine at $\sigma_{\text{KO}} = 0.004$ is -6.93% and -6.95% at $N = 32$ and $N = 40$, i.e. the error does not converge in this range and would be reported as a healthy result by any pipeline that did not compare to a reference. Raising the mask’s dissipation to $\sigma_{\text{KO}} = 0.01$ does not repair it but hides it: the peak becomes ill-defined, the two estimators disagreeing by 1 – 2% (Fig. 2, open symbols), so the masked engine’s answer depends on σ_{KO} at the seven-percent level while the cut-cell answer moves by $\pm 0.25\%$.

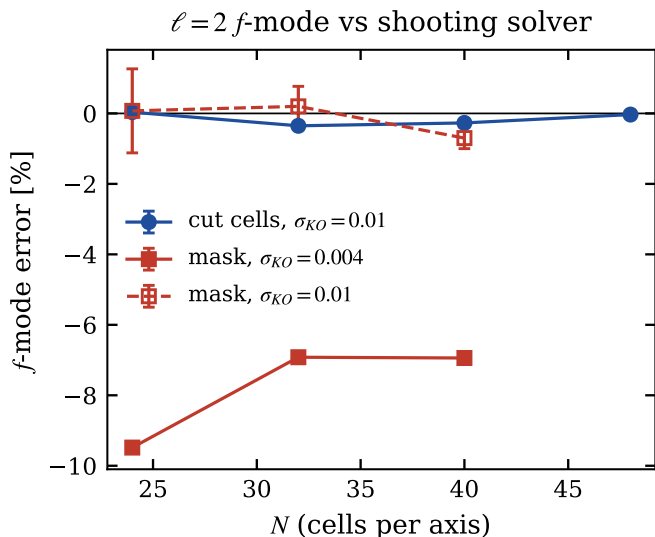


FIG. 2. $\ell = 2$ f -mode of the benchmark star against the shooting solver (1.88291 kHz), as a function of resolution. Filled circles: cut cells, $\sigma_{\text{KO}} = 0.01$. Filled squares: binary mask (rigid wall), $\sigma_{\text{KO}} = 0.004$. Open squares: the mask at $\sigma_{\text{KO}} = 0.01$, where the peak is ill-defined. Symbols are the mean of the periodogram and matrix-pencil estimates; error bars are half their difference. Cut-cell and $\sigma_{\text{KO}} = 0.01$ mask records are nine periods; the $\sigma_{\text{KO}} = 0.004$ mask records are 23 periods.

TABLE I. Benchmark $\Gamma = 2$ polytrope, $\ell = 2$, $m = 0$, $T = 2500 M_{\odot}$ (23 periods, $\Delta f/f = 4.4\%$), $\kappa_{\text{min}} = 0.5$. Reference: shooting solver, $f = 1.88291$, $p_1 = 4.10672$ kHz. Periodogram values; matrix pencil agrees to $\leq 0.13\%$ in every row.

surface	N	σ_{KO}	mode	f [kHz]	dev
cut	24	0.01	f	1.88301	+0.005%
cut	32	0.01	f	1.87560	-0.388%
cut	40	0.01	f	1.87701	-0.313%
cut	48	0.01	f	1.88135	-0.083%
cut	48	0.01	p_1	4.0665	-0.98%
cut	48	0.02	f	1.89076	+0.417%
masked	32	0.004	f	1.75235	-6.93%
masked	40	0.004	f	1.75205	-6.95%

The cut-cell sequence $N = 24/32/40/48$ is non-monotone (+0.005% / -0.39% / -0.31% / -0.08%), so no convergence order can be fitted, Richardson extrapolation on (40,48) makes the answer *worse* (+0.44%), and the honest error bar is the N -spread of $\pm 0.4\%$ rather than the $N = 48$ residual. The Kreiss–Oliger coefficient is a systematic of the same size: $\sigma_{\text{KO}} = 0.02$ gives +0.42% at $N = 48$, bracketing the reference from above, and any cut-cell frequency must state σ_{KO} ; on the masked path it is worth more than seven points. p_2 (5.771 vs 6.031 kHz) is not resolved at these record lengths.

The literature target is the *linear* Cowling column of Font *et al.* [4] (Table 3, Model 2), 1.8825 kHz, which our shooting solver reproduces to 0.02%. Font’s same table

lists a nonlinear HRSC evolution on a fixed spacetime at 1.852 kHz; the 1.7% gap is a finite-amplitude effect our linearized code must not reproduce, and the full-GR value for this star, 1.578–1.586 kHz [5, 6], is 19% below Cowling and is not a valid comparison for a frozen-metric engine.

B. Non-axisymmetric seeds on the full cube

Under the three reflections $x \rightarrow -x$, $y \rightarrow -y$, $z \rightarrow -z$ of the cubic group the $\ell = 2$ quintet splits into the irreps $E_g = \{Y_{20}, \text{Re } Y_{22}\}$ (even under all three) and $T_{2g} = \{\text{Re } Y_{21}, \text{Im } Y_{21}, \text{Im } Y_{22}\}$ (odd under two). Two statements follow for any discretization that respects the cube’s symmetry: cross-moments between irreps vanish identically, and the two E_g partners cannot be split at all. Both are exact tests of the discretization rather than of the physics, and both are met. Seeding Y_{20} , $\text{Re } Y_{22}$ and $\text{Re } Y_{21}$ in turn at $N = 24$ and recording all three moments in each run, every cross-moment is below 10^{-8} of the seeded one, and the E_g partners give $f = 1.88460350153321$ and 1.88460350153323 kHz—identical to fourteen figures. With the previous 24-point Gauss geometry the same pair differed at 5×10^{-7} and the cross-moments were 10^{-5} : the asymmetric κ was the entire effect. The T_{2g} member, by contrast, is genuinely split from E_g by the grid: $f(\text{Re } Y_{21}) = 1.8633$ kHz, -1.13%, at $\Delta x/R = 0.10$. This is the cubic anisotropy of the second-order stencil made visible as a mode splitting. It decreases with resolution—1.12, 0.40 and 0.32% at $N = 24, 32, 40$ (Fig. 3)—consistent with $O(h^2)$ within the $\pm 0.1\%$ spread of the two estimators on nine-period records, while the E_g splitting stays below $10^{-12}\%$ at every N . It is a cleaner measurement of the grid’s anisotropy than the nonlinear engine could give (Sec. VI).

C. Across equations of state

The background builder accepts any barotrope, and the Read *et al.* piecewise polytropes SLy, APR4, H4 and MS1 [8] reproduce their 1D masses and radii to six figures on the grid. Table II shows the outcome. The three smooth analytic barotropes converge to $\leq 0.08\%$ at $N = 48$, all dipping to -0.3 to -0.7% at intermediate N first. The four crusted EOS behave differently: at $\sigma_{\text{KO}} = 0.01$ a surface mode grows at +0.7 to +1.6 e-folds per f -period for $N \geq 40$ (end/start amplitude ratios 4×10^{11} to 8×10^{14} over 23 periods), so they are usable only at $N \leq 32$, at -1.6 to -4%; Richardson on the monotone (24,32) pair gives -0.5% (APR4) to -2.6% (H4).

What discriminates the two classes is neither compactness, stiffness nor tabulation but the steepness of the surface. Writing $\varepsilon \propto (R - r)^n$ between $0.99R$ and $0.999R$, the effective index is $n = 0.50, 0.67, 1.005$ for $\Gamma = 3, 2.5, 2$ (stable) and 1.84, 1.87, 1.95, 1.95 for MS1, H4, APR4,

TABLE II. Cut-cell f -mode across EOS at $M \simeq 1.4 M_\odot$, $\sigma_{\text{KO}} = 0.01$ unless stated, against the shooting value for the identical star. R = Richardson on the monotone (24,32) pair.

EOS ($M/M_\odot$, R/km)	N	f_{3D}	f_{1D}	dev
$\Gamma = 2$ (1.400, 14.15)	48	1.88135	1.88291	-0.083%
$\Gamma = 2.5$	48	2.23448	2.23404	+0.020%
$\Gamma = 3$	48	2.40545	2.40739	-0.080%
SLy (1.421, 11.67)	32	2.37154	2.41928	-1.97%; R -1.02%
	40, 48	VOID		surface mode
	48, $\sigma_{\text{KO}}=0.04$	2.31598		-4.27% (biased)
APR4 (1.360, 11.31)	32	2.45373	2.49463	-1.64%; R -0.51%
	40, 48	VOID		end/start 8×10^{14}
H4 (1.406, 13.93)	32	1.89169	1.94832	-2.91%; R -2.57%
MS1 (1.359, 14.90)	24	1.69672	1.76756	-4.01% (only stable N)
	40, 48, $\sigma_{\text{KO}}=0.04$	1.646, 1.643		-6.9, -7.1%

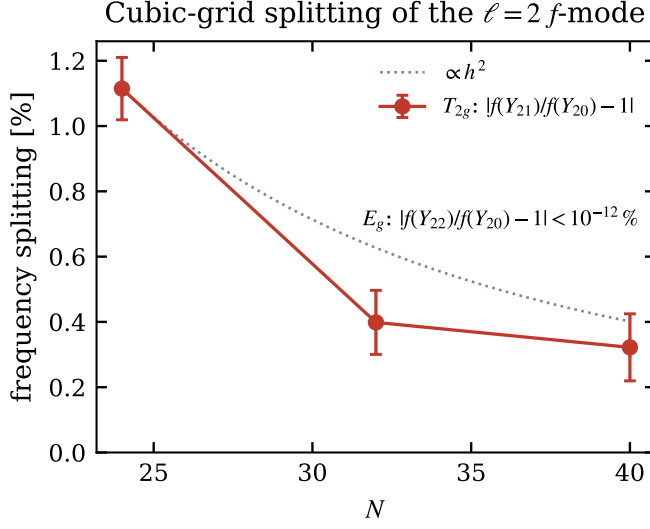


FIG. 3. Splitting of the $\ell = 2$ f -mode by the Cartesian grid. The T_{2g} member $\text{Re } Y_{21}$ is split from the E_g pair by the cubic anisotropy of the stencil; the dotted line is h^2 scaling from $N = 24$. The E_g partners Y_{20} and $\text{Re } Y_{22}$ are degenerate to round-off at every resolution.

SLy (unstable), and the ordering in n is the ordering in the $N = 48$ end/start ratio, monotone over nineteen decades with no exception (Fig. 4). The SLy crust's outermost segment has $\Gamma \simeq 1.58$, hence $n = 1/(\Gamma - 1) \simeq 1.7$ –2.0, and $\varepsilon_0 + p_0$ falls by a factor ~ 500 across one grid cell exactly where the apertures are partial, against 0.44–0.70 for the analytic EOS. This is a correlation across seven EOS, not a demonstrated mechanism; the one causal hypothesis tested—that the finite atmosphere sound speed of the tabulated EOS ($c_s^2 \sim 3 \times 10^{-5}$ versus 10^{-10} – 10^{-19}) drives it—was refuted directly by zeroing c_s^2 in all 77 040 atmosphere cells of the SLy $N = 48$ run: growth +0.596 versus +0.595 e-folds per period. Raising σ_{KO} to 0.04 stabilizes every crusted run for the full 23 periods but

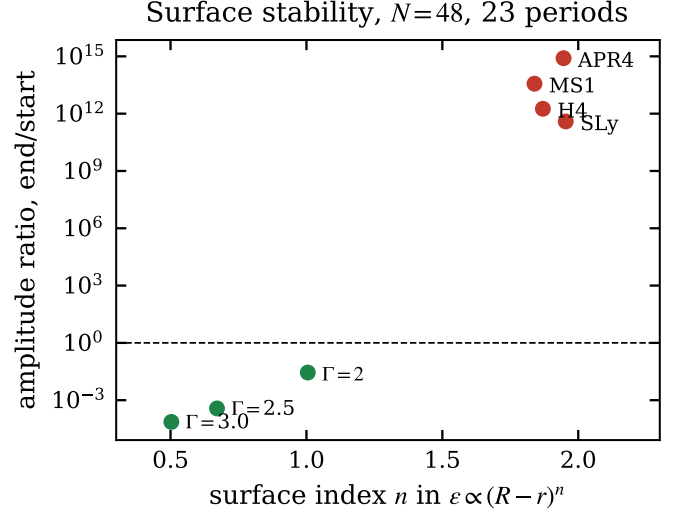


FIG. 4. Growth or decay of the f -mode record over 23 periods at $N = 48$, against the surface index n of each equation of state ($\varepsilon \propto (R - r)^n$ near $r = R$). Green: the three analytic polytropes, decaying. Red: the four crusted Read *et al.* EOS, growing by 10^{11} – 10^{15} . The dashed line is neutral stability.

biases f by -3.8 to -7.1%, with the bias *growing* with N ; there is no configuration we could afford that gives a converging crusted-EOS f -mode, and the mechanism is open.

V. CAUSAL VISCOUS RESULTS

A. Frame robustness of the damped f -mode

Table III gives the $\ell = 2$ f -mode at fixed physical viscosity $\hat{\eta} = 0.03$, $\zeta = 0$, in the three Clarisse frames, with the $\hat{\eta} = 0$ control and the difference protocol. The frame times under Eq. (8) are nearly uniform across the star because $\eta \propto \varepsilon_0$: $\tau_\varepsilon \in [0.168, 0.188]$, $[0.337, 0.375]$,

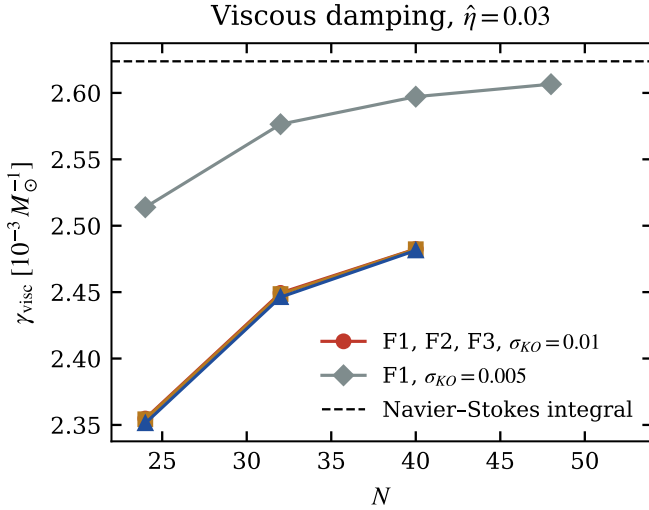


FIG. 5. Viscous damping rate of the f -mode at $\hat{\eta} = 0.03$ from the difference protocol, against resolution. The three Clarisse frames at $\sigma_{\text{KO}} = 0.01$ lie on one curve (spread 0.045% at $N = 40$). At $\sigma_{\text{KO}} = 0.005$ (frame F_1 , matrix pencil) the rate approaches the Navier–Stokes dissipation integral for this star (dashed).

$[0.674, 0.750] M_{\odot}$ for $F_{1,2,3}$.

The frame spread of the damping rate shrinks with resolution, $0.150 \rightarrow 0.130 \rightarrow 0.045\%$ at $N = 24/32/40$, the signature of a frame-invariant physical rate whose apparent spread is discretization residue. The rate itself converges at observed order ≈ 3 ($2.35482 \rightarrow 2.44941 \rightarrow 2.48251 \times 10^{-3}$; successive differences in ratio 2.86), and Richardson extrapolation gives $2.51\text{--}2.54 \times 10^{-3} M_{\odot}^{-1}$. That sequence is at $\sigma_{\text{KO}} = 0.01$; at $\sigma_{\text{KO}} = 0.005$ the matrix-pencil rate is 2.597×10^{-3} at $N = 40$ and 2.607×10^{-3} at $N = 48$ —99.2% of the Newtonian-form Navier–Stokes dissipation integral $0.08746 \hat{\eta} = 2.6238 \times 10^{-3}$ for this star (Fig. 5). The apparent few-percent “causal suppression” of the $\sigma_{\text{KO}} = 0.01$ sequence was Kreiss–Oliger bias. The frame spread of the frequency, $< 0.05\%$, is an upper bound only ($\Delta f/f = 4.4\%$), but is $25\times$ smaller than the 1.22% of the 1D polar operator; that operator’s own damping rates were withheld after its velocity recovery was found to omit the lapse, and the 3D engine—which carries the lapse in Eq. (6)—independently confirms the retraction. The viscous frequency shift relative to the inviscid control is the one quantity that required a dedicated convergence study (Sec. V C).

B. a_1 is inert

Holding $a_2 = 25$ and sweeping $a_1 = 6.25/25/100$ gives $f = 1.89112/1.89113/1.89123$ kHz (spread 0.006%) and $\gamma = 5.618/5.621/5.630 \times 10^{-3}$ (spread 0.22%) over a $16\times$ range. The three Clarisse frames differ in the stellar polar sector through a_2 alone. In the axial sector this is exact— $\mathcal{A}$ is a scalar and odd-parity perturbations carry no scalar

content—and here it is empirical.

C. The viscous frequency shift

At $\sigma_{\text{KO}} = 0.01$ the shift $\Delta f/f = f(\hat{\eta})/f(0) - 1$ measured $+0.40/+0.28/+0.15\%$ at $N = 24/32/40$ with non-shrinking differences, and could only be quoted as an upper bound. Two observations settled it (33 runs; `repro/bdnk3d-viscous_shift.jl`; Fig. 6). First, with the frame times held fixed and $\hat{\eta}$ swept, the shift is roughly linear in $\hat{\eta}$, whereas every physical mechanism—the damped-oscillator correction and the reactive frame terms—is second order. Second, halving σ_{KO} removes $\approx 0.35\%$ of the shift at every N : a physical shift cannot depend on the dissipation coefficient, so the positive shift was a Kreiss–Oliger artifact. What survives at $\sigma_{\text{KO}} = 0.005$ converges to the damped-oscillator value, which has no free parameter because both damping rates are measured in the same runs:

$$\frac{\Delta f}{f} = -\frac{\gamma_{\text{tot}}^2 - \gamma_0^2}{2\omega^2}. \quad (11)$$

At $\hat{\eta} = 0.03$ the measured shift (mean of the two estimators) against Eq. (11) is $+0.05/-0.20\%$, $-0.03/-0.16\%$, $-0.08/-0.13\%$ and $-0.125/-0.133\%$ at $N = 24/32/40/48$; the residual $+0.25 \rightarrow +0.13 \rightarrow +0.05 \rightarrow +0.01\%$ falls roughly as h^3 , and at $N = 40$ the $\hat{\eta}$ -dependence is quadratic once that offset is removed ($\hat{\eta} = 0.015, 0.03, 0.045$). Frame-reactive corrections are $O(\tau^2 \omega^2) \approx 3 \times 10^{-5}$, below resolution. The converged statement is $\Delta f/f(\hat{\eta}=0.03) = -0.125 \pm 0.013\%$, the damped-oscillator shift. Three protocol facts the data enforce: the frame must be held fixed while $\hat{\eta}$ is swept, or the η -dependence is entangled with $\tau \propto \hat{\eta}$; the inviscid control becomes unstable below $\sigma_{\text{KO}} \approx 0.005$ at $N = 24$ and 32 (the cut-cell surface mode) while viscous runs survive to 0.00125 , so $\sigma_{\text{KO}} = 0.005$ is the floor of the difference protocol and is set by the control; and the compact viscous stencil (`visc_compact`) shifts f by $+0.15\%$ and cuts γ_{visc} by 32% at $N = 24$ and is not to be used for the mode.

D. The stability bound is sufficient, not necessary

F_1 violates Eq. (7) in 100% of cells and 100% of proper volume, yet evolves stably for 46 periods through twelve decades of decay with a window-by-window decay rate constant to 1%; it is as clean as F_3 , which violates it in 51%. The local sound-sector analysis explains why. The unstable band lies above a critical wavenumber $k_c = C/\hat{\eta}$ with $C = 0.4599, 0.3019, 0.1753$ for $F_{1,2,3}$ (minimum over the star, always at $r/R = 0.994$), i.e. $k_c = 15.3, 10.1, 5.8 M_{\odot}^{-1}$ at $\hat{\eta} = 0.03$. The second-order centred stencil caps the effective wavenumber at $k_{\text{eff}} = \sin(kh)/h \leq 1/h = N/23.0$, so the unstable band is resolved only for $N > 353, 232, 135$ —beyond reach

TABLE III. BDNK f -mode in the three frames, $\hat{\eta} = 0.03$, cut cells, $\sigma_{KO} = 0.01$, $N = 40$, $T = 2500 M_\odot$ (22.8 periods). $\gamma_{\text{visc}} = \gamma(\hat{\eta}) - \gamma(0)$; 1D polar values from the ℓ -reduced frequency-domain operator with the corrected free-surface BC.

	F_1	F_2	F_3	spread
(a_1, a_2)	(25/4, 25/7)	(25/2, 25/3)	(25, 25)	
f pencil [kHz]	1.87954	1.87973	1.88040	0.046%
f periodogram	1.88027	1.88036	1.88082	0.029%
f , 1D polar	1.9038	1.9105	1.9272	1.22%
f , $\hat{\eta} = 0$ control	1.87681	1.87681	1.87681	0
$\gamma_{\text{visc}} [10^{-3} M_\odot^{-1}]$	2.48251	2.48229	2.48139	0.045%
τ_{damp}, Q	1.984 ms, 11.72	1.984 ms, 11.72	1.985 ms, 11.73	
cells violating (7)	100%	94.4%	51.3%	
stable, 22.8 periods	yes	yes	yes	

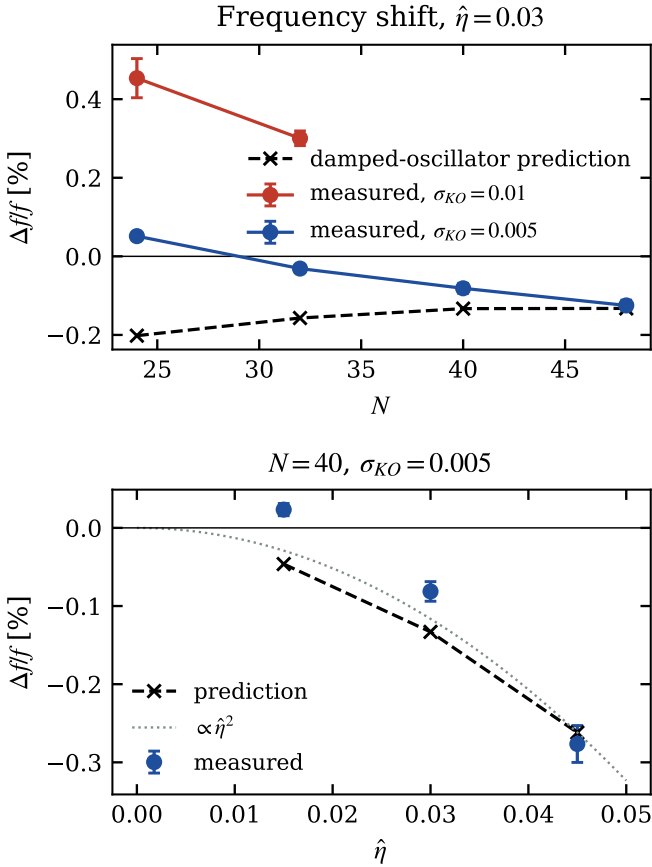


FIG. 6. Viscous frequency shift of the f -mode in frame F_1 . Top: against resolution at $\hat{\eta} = 0.03$, for two Kreiss–Oliger coefficients, with the damped-oscillator prediction $-(\gamma^2 - \gamma_0^2)/2\omega^2$ built from the measured damping rates (crosses). Bottom: against $\hat{\eta}$ at $N = 40$, $\sigma_{KO} = 0.005$; the dotted curve is $\propto \hat{\eta}^2$. Symbols are the mean of the two estimators, error bars half their difference.

at $\sim N^4$ cost. The criterion “unstable iff $k_c < 1/h$ ” was therefore tested on the cheap $\hat{\eta}$ axis at $N = 24$ ($1/h = 1.043$; Fig. 7): six predictions, six correct in sign, with growth rates agreeing to 24–28% where growth is

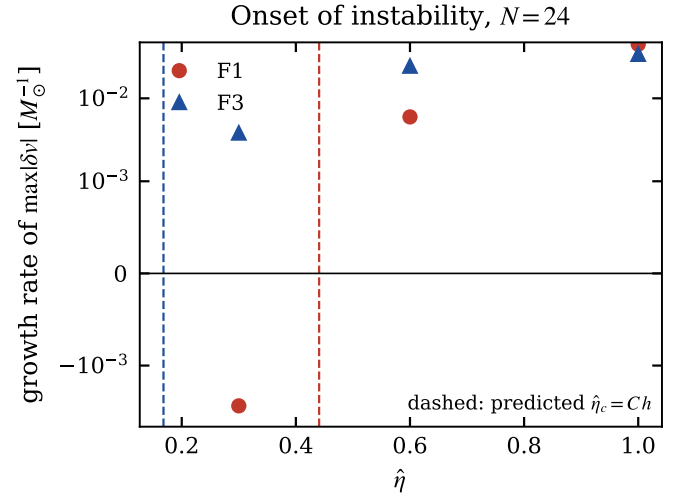


FIG. 7. Measured growth rate of $\max|\delta v|$ against viscosity at $N = 24$ for frames F_1 (circles) and F_3 (triangles), with the frame times built from each run’s own $\hat{\eta}$. Dashed lines: the predicted onset $\hat{\eta}_c = Ch$ from the critical-wavenumber criterion. F_3 destabilises first although it violates the stability bound in fewer cells.

well developed (F_1 , $\hat{\eta} = 1.0$: predicted 3.50×10^{-2} , measured 4.49×10^{-2}). And the ordering inverts: at $\hat{\eta} = 0.60$, F_3 blows up (q_{20} grows 5×10^4) while F_1 decays, because $k_c(F_3)$ is $2.6\times$ smaller. *The violation fraction does not order the instability; the critical wavenumber does.* The published fractions 60.7% and 21.4% from the 1D radial grid for F_2, F_3 are a counting artifact; the 3D cell fractions match the 1D proper-volume fractions (94.6%, 52.6%) to $< 1\%$.

VI. COMPARISON WITH A NONLINEAR DG ENGINE

The same problem was run in a full 3+1D nodal Runge–Kutta discontinuous-Galerkin Valencia GRHydro code on the frozen metric (densitized conserved variables,

directional Rusanov sweeps, SSP-RK3, TVB/Zhang–Shu limiter, atmosphere, well-balanced background subtraction), on an octant grid with a full 3-momentum. It is the only engine here that can carry a non-axisymmetric perturbation, and it does: a $\text{Re} Y_{22} \propto n_x^2 - n_y^2$ seed is detected by the $m = 2$ quadrupole with contrast $3 \times 10^5 - 5 \times 10^6$ over the $m = 0$ seed at every resolution. On a non-rotating star the m -degeneracy is exact, so the measured $|f(m=2) - f(m=0)|/f$ of 6.4/3.8/2.1/0.81% at $K = 5/6/8/10$ elements per axis is a grid artifact falling as $h^{3.2}$, consistent with the scheme’s $O(h^3)$; a single run seeded with Y_{21} , whose mirror extension populates both E_g channels on one background, bounds the grid’s genuine degeneracy violation at 9×10^{-5} —a bound that the linear engine’s fourteen-figure E_g agreement (Sec. IV B) has since superseded by ten orders of magnitude. Two facts, however, make the linear cut-cell method the one to use for precision. The DG background is not static: the well-balanced subtraction is exact but the limiter is not, and ρ_c erodes -10% by 4.6 periods and -31 to -43% by $3000 M_\odot$, capping the usable record at $\Delta f/f = 21.6\%$ and putting its absolute f at -16% . And the octant’s three separate reflections admit only the E_g doublet $\{Y_{20}, \text{Re} Y_{22}\}$; the entire T_{2g} triplet, including all of $m = 1$, is structurally absent from the domain. A search for nonlinear amplitude dependence in the DG engine returned no resolved shift: the one suggestive point ($-1.4 \pm 1.0\%$ at seed amplitude 0.1, $K = 8$) becomes $-7.5 \pm 0.9\%$ at $K = 6$, a $5.4\times$ change for a 25% refinement, and no second harmonic is detected ($A_{2f}/A_f < 0.09\text{--}0.19$ at $\xi/R = 0.41$). The linear engine’s exact amplitude proportionality was the control that established this.

VII. LIMITATIONS AND DEFECTS

Structural. The metric is frozen: no gravitational-wave damping, no w -modes, no tidal deformability. The engine is linear. Its public interface is now a single call, `bdnk3d.qnm`, which builds the star and engine, installs a Clarisse frame from (a_1, a_2) or a scalar $\hat{\tau}$, seeds any (ℓ, m) , evolves, applies the gated two-estimator read-out and the $\hat{\eta} = 0$ difference protocol, and returns the frequency, damping, floor, Rayleigh resolution, stability flag and bound-violation fraction together; `test/test.bdnk3d.qnm.jl` verifies each of these against the numbers in this paper at $N = 24$.

Numerical. Time-domain resolution at affordable cost is $\Delta f/f = 4.4\%$; σ_{KO} is a $\pm 0.25\%$ systematic on absolute frequencies and, at $\sigma_{\text{KO}} = 0.01$, produced a spuri-

ous $+0.3\%$ viscous frequency shift (Sec. V C); the cut-cell path develops a late-time growing surface mode at $\sigma_{\text{KO}} = 0.004$ for $N \geq 32$ (growth 3×10^{-4} to $9 \times 10^{-3} M_\odot^{-1}$ from $N = 32$ to 56), cured at $\sigma_{\text{KO}} = 0.01$ through $N = 48$ but unverified above; realistic crusted EOS are limited to $N \leq 32$; the numerical damping floor $\gamma(0) \approx 1.2 \times 10^{-3} M_\odot^{-1}$ at $N = 48$ is a hard floor under any viscous measurement without the difference protocol; and p_2 is not resolved.

Defects found in the code, all recorded rather than silently fixed. The `setup_evo3d` default is `cut=false`, i.e. the -6.95% masked engine, with no symptom. The background builder sets $\rho_0 = \varepsilon - p$, exact only for $\Gamma = 2$; harmless to these linear engines, which never read ρ_0 , but fatal to any nonlinear user. The ℓ -reduced 1D polar operator’s velocity recovery omits the lapse. And a Julia lexical trap: `2f1` parses as the `Float32` literal 20.0, not $2f_1$, which turned a harmonic search into a fake null until a synthetic-injection test caught it.

VIII. CONCLUSIONS

A linearized Cowling evolution on a non-conforming Cartesian grid reproduces the frequency-domain f -mode of smooth relativistic polytropes to better than 0.1% once two things are right that are easy to get wrong: the lapse on the Euler equation, and the boundary condition at the stellar surface. The first is a derivation. The second is a cut-cell finite-volume divergence from which the boundary face is simply absent—which imposes a free surface at the true sphere instead of a rigid wall on the staircase, and is worth seven percent. Extended to BDNK with the frame times kept and the recovery written in proper time, the engine gives a viscous damping rate that is frame-invariant to 0.045%, converges at third order to within a few percent of the Navier–Stokes dissipation integral, and exposes the linear stability bound of the theory as sufficient but not necessary on a star, with the critical wavenumber rather than the violation fraction controlling onset. The crusted-EOS surface instability is the open problem; its discriminant is known and its mechanism is not.

ACKNOWLEDGMENTS

The validation runs, the frame study and the DG comparison were executed as adversarially verified agent workflows: every headline number was re-run by an independent verifier, and the verifiers’ retractions are incorporated above.

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